Priyam Gupta

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Profile

I am a PhD researcher specialising in **scientific machine learning**, computational modelling, and numerical methods for complex non-linear dynamical systems. My research develops **data-driven surrogate models** for high-dimensional simulation data, combining modern state-space models, non-linear dimensionality reduction, and statistical physics. I design and train models in **Python/PyTorch** and integrate them with numerical simulation, optimisation, and control pipelines in Linux/HPC environments. My work spans fluid dynamics, stochastic simulation, reduced-order modelling, and adaptive control.

Education

Imperial College London: PhD, Department of Aeronautics — 11/2022–10/2026
Topic: Computational modelling, numerical methods, and control of non-linear dynamical systems — London, UK

Imperial College London: MSc, Aeronautics (Advanced Computational Methods) — 10/2021–09/2022
Result: **Distinction**

Delhi Technological University (formerly DCE): B.Tech., Mechanical Engineering — 2017–2021
CGPA: 8.78/10 (Distinction) — New Delhi, India

Experience

Imperial College London, *Research Assistant* — **Jun 2026 - Present**

- Developing **Monte Carlo simulation methods** for stochastic modelling of diffusion processes in complex geometries, with application to cardiac MRI.
- Building computational workflows for ensemble particle simulations, including trajectory generation, statistical analysis, and quantitative model validation.

Imperial College London, *PhD Researcher* — **Nov 2022 - Oct 2026**
Supervisors: Dr Georgios Rigas, Dr Denis Sipp, Dr Taraneh Sayadi, Dr Peter J. Schmid

- Developed **state-space and Koopman-based surrogate models** for forecasting high-dimensional non-linear flow dynamics from simulation and partially observed data (published in Proc. Royal Soc. A).
- Built modular **Python/PyTorch** research frameworks integrating simulation-data processing, model training, multi-step evaluation, and numerical experiments on Linux/HPC systems.
- Designed and implemented adaptive **model predictive control** pipelines, integrating learned surrogate models with numerical optimisation for control of chaotic fluid systems.

Imperial College London, *Graduate Teaching Assistant* — **Jan 2023 - Dec 2023**

- Conducted tutorials on computational methods, numerical simulation, and turbulence modelling.

Magri Lab - Imperial College London, *MSc Research Project* — **May 2022 - Aug 2022**

- Developed **physics-constrained non-linear autoencoders** for dimensionality reduction of chaotic dynamical systems, incorporating conservation laws and symmetry constraints into the latent representation.

Monolith AI - London, *Research Intern* — **Jun 2020 - Sep 2020**

- Built end-to-end **simulation-to-ML pipelines** for surrogate modelling of 2D/3D flow fields, from data generation with **OpenFOAM** to model training and evaluation.
- Developed machine-learning models for engineering applications including airfoils, wind turbines, and re-entry vehicles.
- Designed custom loss functions targeting critical engineering design regions, improving predictive accuracy in high-consequence operating regimes.

Delhi Technological University, *Research Assistant* — **Jul 2018 - Jul 2021**
Supervisor: Prof. Raj Kumar Singh

- Developed a data-driven inverse airfoil design framework using deep convolutional generative models.
- Conducted quantitative benchmarking of regression methods for pressure-field reconstruction from PIV data.
- Built a shape optimisation framework using an invasive-weed-inspired genetic algorithm for slat-airfoil configurations.

Selected Publications

- **Gupta, P.**, Schmid P., Sipp D., Sayadi T., Rigas G., *"Non-Markovian Bilinear Representation for Non-Linear Actuated Flows Using Structured State-Space Models," Journal of Fluid Mechanics*, manuscript in preparation.
- **Gupta, P.**, Schmid P., Sipp D., Sayadi T., Rigas G., *"Mori–Zwanzig Latent-Space Koopman Closure for Non-Linear Autoencoders," Proceedings of the Royal Society A*, (2025). doi.org/10.1098/rspa.2024.0259
- **Gupta, P.**, Tyagi, P., Singh, R. K., *"Analysis of Generative Adversarial Networks for Data-Driven Inverse Airfoil Design,"* Lecture Notes in Networks and Systems, Springer, (2022). doi.org/10.1007/978-981-16-7618-5_22

Technical Skills

Programming: Python, C++, MATLAB, CUDA
Machine Learning: PyTorch, State-Space Models, Autoencoders, Reinforcement Learning
Simulation & Scientific Computing: NumPy, SciPy, OpenFOAM, FEniCS, CasADi, Monte Carlo Methods
Core Areas: Surrogate Modelling, Reduced-Order Modelling, Time-Series Modelling, Numerical Optimisation, Model Predictive Control